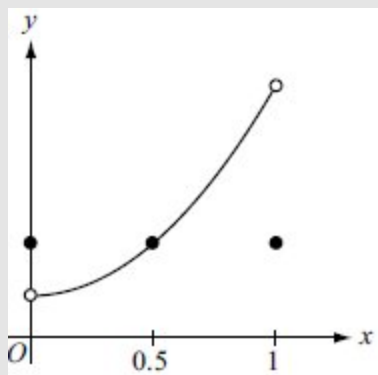


5-2-

1. The function f is defined on the closed interval $[0, 1]$ and satisfies $f(0) = f\left(\frac{1}{2}\right) = f(1)$. On the open interval $(0, 1)$, f is continuous and strictly increasing. Which of the following statements is true?
- (A) f attains both a minimum value and a maximum value on the closed interval $[0, 1]$.
- (B) f attains a minimum value but not a maximum value on the closed interval $[0, 1]$.
- (C) f attains a maximum value but not a minimum value on the closed interval $[0, 1]$.
- (D) f attains neither a minimum value nor a maximum value on the closed interval $[0, 1]$. ✓

Answer D

This option is correct. The graph of a function that satisfies the conditions in the stem would have to behave something like the figure below.



The graph on the open interval $(0, 1)$ must be strictly increasing and the three solid dots must line up horizontally. The solid dots and the open dots at $x = 0$ and $x = 1$ cannot coincide. A function with such a graph has neither a minimum value nor a maximum value on the closed interval $[0, 1]$. Note that the Extreme Value Theorem does not apply here because f is *not* continuous on $[0, 1]$.

A more detailed explanation is that because f is strictly increasing on the open interval $(0, 1)$, if it attains a minimum value on the closed interval $[0, 1]$ then that value must occur at either $x = 0$ or $x = 1$. But that is impossible because it would lead to the contradiction

$$f\left(\frac{1}{2}\right) = f(a) \leq f\left(\frac{1}{4}\right) < f\left(\frac{1}{2}\right), \text{ where } a \text{ is either } 0 \text{ or } 1.$$

The first inequality comes from $f(a)$ being the minimum value, and the second inequality comes from f being strictly increasing on $(0, 1)$.

Similarly, if the function attains a maximum value on the closed interval $[0, 1]$, that value must also occur at either $x = 0$ or $x = 1$. But that is again impossible because it would lead to the contradiction

$$f\left(\frac{1}{2}\right) < f\left(\frac{3}{4}\right) \leq f(b) = f\left(\frac{1}{2}\right), \text{ where } b \text{ is either } 0 \text{ or } 1.$$

5-2-

The first inequality comes from f being strictly increasing on $(0, 1)$, and the second inequality comes from $f(b)$ being the maximum value.

2. Let f be the function defined by $f(x) = \frac{\ln x}{x}$. What is the absolute maximum value of f ?

(A) 1

(B) $\frac{1}{e}$



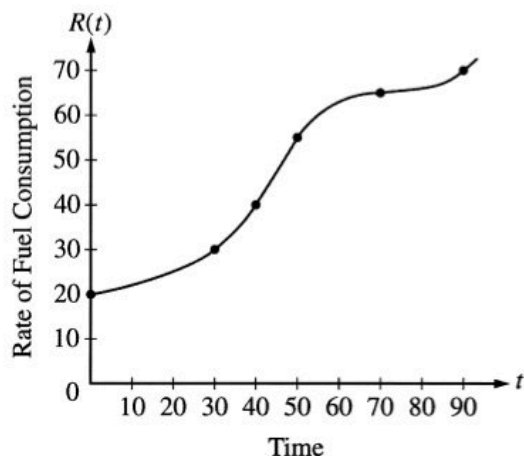
(C) 0

(D) $-e$

(E) f does not have an absolute maximum value.

5-2-

3.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
---	---	---

The student response earns two of the following points:

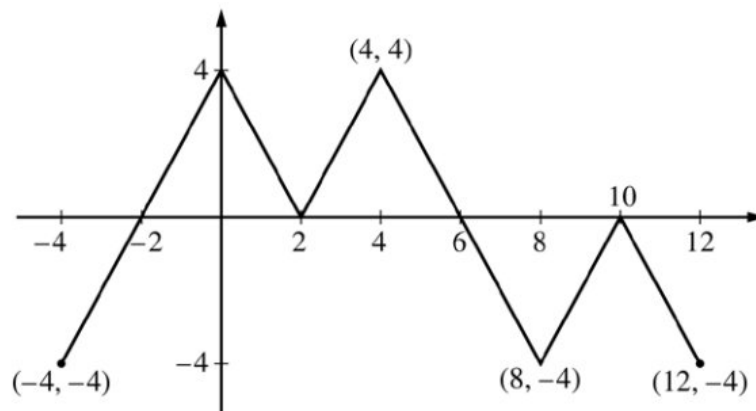
1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

5-2-

4.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_2^x f(t) dt$.

- Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.
- Does the graph of g have a point of inflection at $x = 4$? Justify your answer.
- Find the absolute minimum value and the absolute maximum value of g on the interval $-4 \leq x \leq 12$. Justify your answers.
- For $-4 \leq x \leq 12$, find all intervals for which $g(x) \leq 0$.

Part A

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

Part B

5-2-

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for answer with justification

The graph of g has a point of inflection at $x = 4$ since $g'(x) = f(x)$ is increasing for $2 \leq x \leq 4$ and decreasing for $4 \leq x \leq 8$.

Part C

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.



0	1	2	3	4	5
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5-2-

The student response earns all of the following points:

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for considering $x = -2$ and $x = 6$ as candidates

$g'(x) = f(x)$ changes sign only at $x = -2$ and $x = 6$

1 point is earned for considering $x = -4$ and $x = 12$

x	$g(x)$
-4	-4
-2	-8
6	8
12	-4

2 points are earned for answers with justification

On the interval $-4 \leq x \leq 12$, the absolute minimum value is $g(-2) = -8$ and the absolute maximum value is $g(6) = 8$.

Part D

1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

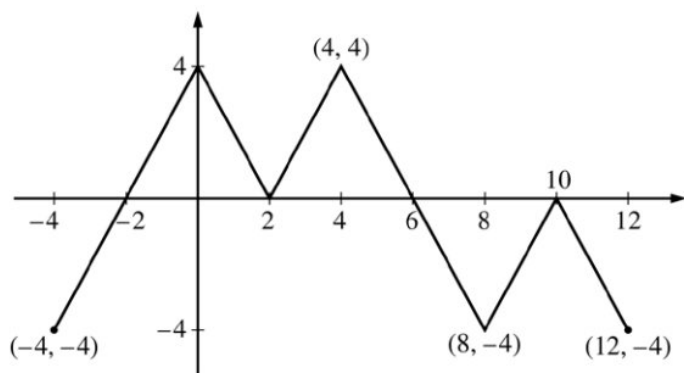
1 point is earned for $g'(x) = f(x)$ in (a), (b), (c), or (d)

2 point are earned for intervals

$g(x) \leq 0$ for $-4 \leq x \leq 2$ and $10 \leq x \leq 12$.

5-2-

5.

Graph of f

The figure above shows the graph of the piecewise-linear function f . For $-4 \leq x \leq 12$, the function g is defined by $g(x) = \int_{-4}^x f(t) dt$.

Does g have a relative minimum, a relative maximum, or neither at $x = 10$? Justify your answer.

Part A

1 point is earned for: $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for: answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.



0	1	2
---	---	---

Student response earns 2 of the following 2 point(s)

1 point is earned for: $g'(x) = f(x)$ in (a), (b), (c), or (d)

1 point is earned for: answer with justification

The function g has neither a relative minimum nor a relative maximum at $x = 10$ since $g'(x) = f(x)$ and $f(x) \leq 0$ for $8 \leq x \leq 12$.

5-2-

6. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(x) = \begin{cases} \sqrt{9-x^2} & \text{for } -3 \leq x \leq 0 \\ -x + 3\cos\left(\frac{\pi x}{2}\right) & \text{for } 0 < x \leq 4 \end{cases}$$

Let f be the function defined above.

- Find the average rate of change of f on the interval $-3 \leq x \leq 4$.
- Write an equation for the line tangent to the graph of f at $x = 3$.
- Find the average value of f on the interval $-3 \leq x \leq 4$.
- Must there be a value of x at which $f(x)$ attains an absolute maximum on the closed interval $-3 \leq x \leq 4$? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.



0	1
---	---

The student response accurately includes the criteria below.

☐ answer

Solution:

$$\text{Average rate of change} = \frac{f(4) - f(-3)}{4 - (-3)} = \frac{-1 - 0}{7} = -\frac{1}{7}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-2-

The first point is earned with a correct and labeled value of $f'(3)$.

The second point can be earned with a consistent tangent line equation created from an incorrect value of $f'(3)$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $f'(3)$
- ☐ equation

Solution:

$$f(3) = -3 + 3 \cos\left(\frac{3\pi}{2}\right) = -3$$

$$\text{For } 0 < x < 4, f'(x) = -1 + \left(-3 \sin\left(\frac{\pi x}{2}\right)\right) \cdot \frac{\pi}{2}$$

$$f'(3) = -1 + \left(-3 \sin\left(\frac{3\pi}{2}\right)\right) \cdot \frac{\pi}{2} = -1 + \frac{3\pi}{2}$$

$$\text{An equation for the tangent line is } y = -3 + \left(-1 + \frac{3\pi}{2}\right)(x - 3).$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The constant $\frac{1}{b-a}$ is part of the fourth point.

Simplification is not required for the second point.

The third point is earned with the antiderivative $\frac{-x^2}{2} + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)$. The third point can also be earned with $\frac{-x^2}{2} + \frac{2}{\pi} \sin\left(\frac{\pi x}{2}\right)$, though this response will not be eligible for the fourth point.

Simplification is not required for the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ integrals of f over $-3 \leq x \leq 0$ and $0 \leq x \leq 4$

5-2-

- ☐ value of $\int_{-3}^0 \sqrt{9-x^2} \, dx$
- ☐ antiderivative of $-x + 3 \cos\left(\frac{\pi x}{2}\right)$
- ☐ answer

Solution:

The average value of f on the interval $-3 \leq x \leq 4$ is $\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx$.

$$\int_{-3}^4 f(x) \, dx = \int_{-3}^0 f(x) \, dx + \int_0^4 f(x) \, dx$$

$$\int_{-3}^0 f(x) \, dx = \int_{-3}^0 \sqrt{9-x^2} \, dx = \frac{9\pi}{4}$$

$$\int_0^4 f(x) \, dx = \int_0^4 \left(-x + 3 \cos\left(\frac{\pi x}{2}\right)\right) \, dx = \left[-\frac{1}{2}x^2 + \frac{6}{\pi} \sin\left(\frac{\pi x}{2}\right)\right]_0^4 = -8$$

$$\frac{1}{4-(-3)} \int_{-3}^4 f(x) \, dx = \frac{1}{7} \left(\frac{9\pi}{4} - 8\right)$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned by evaluating each branch of the function at $x = 0$ instead of evaluating the one-sided limits, but the response will not be eligible for the second point.

The second point is earned with correct limit notation for the first point, clear communication that $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0) = 3$, and a statement that the function is continuous on the closed interval, and therefore must have an absolute maximum. Specifically naming the Extreme Value Theorem is not required to earn this point. Responses that indicate that the function is differentiable on the interval $[-3, 4]$ will not earn the point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ continuity at $x = 0$
- ☐ answer with justification

5-2-**Solution:**

$\lim_{x \rightarrow 0^-} f(x) = f(0) = 3$ and $\lim_{x \rightarrow 0^+} f(x) = 3$, so f is continuous at $x = 0$.

Because f is continuous on $[-3, 4]$, the Extreme Value Theorem guarantees that f attains an absolute maximum on $[-3, 4]$.

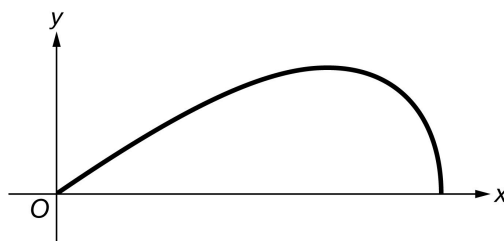
5-2-

7. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



A company designs spinning toys using the family of functions $y = cx\sqrt{4 - x^2}$, where c is a positive constant. The figure above shows the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$, for some c . Each spinning toy is in the shape of the solid generated when such a region is revolved about the x -axis. Both x and y are measured in inches.

(a) Find the area of the region in the first quadrant bounded by the x -axis and the graph of $y = cx\sqrt{4 - x^2}$ for $c = 6$.

(b) It is known that, for $y = cx\sqrt{4 - x^2}$, $\frac{dy}{dx} = \frac{c(4 - 2x^2)}{\sqrt{4 - x^2}}$. For a particular spinning toy, the radius of the largest cross-sectional circular slice is 1.2 inches. What is the value of c for this spinning toy?

(c) For another spinning toy, the volume is 2π cubic inches. What is the value of c for this spinning toy?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Units are not required for any points in this question and are not read if presented (correctly or incorrectly) in any part of the response.

The first point is earned for presenting $cx\sqrt{4 - x^2}$ or $6x\sqrt{4 - x^2}$ as the integrand in a definite integral. Limits of integration (numeric or alphanumeric) must be presented (as part of the definite integral) but do not need to be correct in order to earn the first point.

If an indefinite integral is presented with an integrand of the correct form, the first point can be earned if the antiderivative (correct or incorrect) is eventually evaluated using the correct limits of integration.

The second point can be earned without the first point. The second point is earned for the presentation of a correct antiderivative of a function of the form $Ax\sqrt{4 - x^2}$, for any nonzero constant A . If the response has subsequent

5-2-

errors in simplification of the antiderivative or sign errors, the response will earn the second point but will not earn the third point.

Responses that use u -substitution and have incorrect limits of integration or do not change the limits of integration from x - to u -values are eligible for the second point.

The response is eligible for the third point only if it has earned the second point.

The third point is earned only for the answer 16 or equivalent. In the case where a response only presents an indefinite integral, the use of the correct limits of integration to evaluate the antiderivative must be shown to earn the third point.

The response cannot correct -16 to $+16$ in order to earn the third point; there is no possible reversal here.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Integrand
- ☐ Antiderivative
- ☐ Answer

Solution:

$$6x\sqrt{4-x^2} = 0 \Rightarrow x = 0, x = 2$$

$$\text{Area} = \int_0^2 6x\sqrt{4-x^2} \, dx$$

$$\text{Let } u = 4 - x^2.$$

$$du = -2x \, dx \Rightarrow -\frac{1}{2} du = x \, dx$$

$$x = 0 \Rightarrow u = 4 - 0^2 = 4$$

$$x = 2 \Rightarrow u = 4 - 2^2 = 0$$

$$\begin{aligned} \int_0^2 6x\sqrt{4-x^2} \, dx &= \int_4^0 6 \left(-\frac{1}{2}\right) \sqrt{u} \, du = -3 \int_4^0 u^{1/2} \, du = 3 \int_0^4 u^{1/2} \, du \\ &= 2u^{3/2} \Big|_{u=0}^{u=4} = 2 \cdot 8 = 16 \end{aligned}$$

The area of the region is 16 square inches.

Part B

5-2-

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for setting $\frac{dy}{dx} = 0$, $\frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0$, or $c(4-2x^2) = 0$.

An unsupported $x = \sqrt{2}$ does not earn the first point.

The second point can be earned without the first point but is earned only for the answer $c = 0.6$ with supporting work.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Sets $\frac{dy}{dx} = 0$
- ☐ Answer

Solution:

The cross-sectional circular slice with the largest radius occurs where $cx\sqrt{4-x^2}$ has its maximum on the interval $0 < x < 2$.

$$\frac{dy}{dx} = \frac{c(4-2x^2)}{\sqrt{4-x^2}} = 0 \Rightarrow x = \sqrt{2}$$

$$x = \sqrt{2} \Rightarrow y = c\sqrt{2}\sqrt{4 - (\sqrt{2})^2} = 2c$$

$$2c = 1.2 \Rightarrow c = 0.6$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting an integrand of the form $A(x\sqrt{4-x^2})^2$ in a definite integral with any limits of integration (numeric or alphanumeric) and any nonzero constant A . Mishandling the c will result in the response being ineligible for the fourth point.

The second point can be earned without the first point. The second point is earned for the limits of integration, $x = 0$ and $x = 2$, and the constant π (but not for 2π) as part of an integral with a correct or incorrect integrand.

If an indefinite integral is presented with the correct constant π , the second point can be earned if the antiderivative (correct or incorrect) is evaluated using the correct limits of integration.

5-2-

A response that presents $2 = \int_0^2 \left(cx\sqrt{4-x^2} \right)^2 dx$ earns the first and second points.

The third point is earned for presenting a correct antiderivative of the presented integrand of the form $A\left(x\sqrt{4-x^2}\right)^2$ for any nonzero A . If there are subsequent errors in simplification of the antiderivative, linkage errors, or sign errors, the response will not earn the fourth point.

The fourth point cannot be earned without the third point. The fourth point is earned only for the correct answer. The expression does not need to be simplified to earn the fourth point.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ Form of the integrand
- ☐ Limits and constant
- ☐ Antiderivative
- ☐ Answer

Solution:

$$\begin{aligned}
 \text{Volume} &= \int_0^2 \pi \left(cx\sqrt{4-x^2} \right)^2 dx = \pi c^2 \int_0^2 x^2 (4-x^2) dx \\
 &= \pi c^2 \int_0^2 (4x^2 - x^4) dx = \pi c^2 \left(\frac{4}{3}x^3 - \frac{1}{5}x^5 \right) \Big|_0^2 \\
 &= \pi c^2 \left(\frac{32}{3} - \frac{32}{5} \right) = \frac{64\pi c^2}{15}
 \end{aligned}$$

$$\frac{64\pi c^2}{15} = 2\pi \Rightarrow c^2 = \frac{15}{32} \Rightarrow c = \sqrt{\frac{15}{32}}$$

8. Let g be the function given by $g(x) = x^2 e^{kx}$, where k is a constant. For what value of k does g have a critical point at $x = \frac{2}{3}$?

(A) -3

(B) $-\frac{3}{2}$

(C) $-\frac{1}{3}$

(D) 0

(E) There is no such k .



5-2-

9. How many critical points does the function $f(x) = (x + 2)^5(x - 3)^4$ have?

(A) One

(B) Two

(C) Three

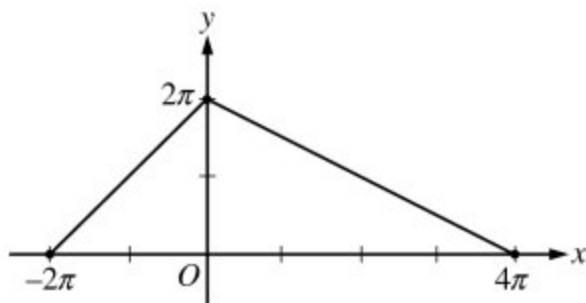


(D) Five

(E) Nine

5-2-

10.

Graph of g

Let g be the piecewise-linear function defined on $[-2\pi, 4\pi]$ whose graph is given above, and let $f(x) = g(x) - \cos\left(\frac{x}{2}\right)$.

Find all x -values in the open interval $(-2\pi, 4\pi)$ for which f has a critical point.

Part B

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x=0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x=0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x=0$ and $x = \pi$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

5-2-

1 point is earned for $x=0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x=0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x)=0$ when $x=\pi$.

f has critical points at $x=0$ and $x=\pi$.

11.  The first derivative of the function f is given by $f'(x) = \frac{\cos^2 x}{x} - \frac{1}{5}$. How many critical values does f have on the open interval $(0,10)$?

(A) One

(B) Three

(C) Four

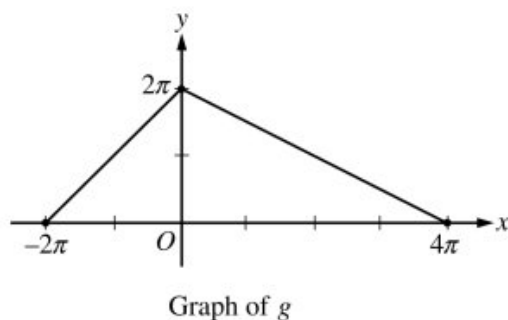
(D) Five

(E) Seven



5-2-

12.



Let g be the piecewise-linear function defined on $[-2\pi, 4\pi]$ whose graph is given above, and let $f(x) = g(x) - \cos\left(\frac{x}{2}\right)$.

- (a) Find $\int_{-2\pi}^{4\pi} f(x) dx$. Show the computations that lead to your answer.
- (b) Find all x -values in the open interval $(-2\pi, 4\pi)$ for which f has a critical point.
- (c) Let $h(x) = \int_0^{3x} g(t) dt$. Find $h'\left(-\frac{\pi}{3}\right)$.

Part A

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \int_{-2\pi}^{4\pi} f(x) dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right) \right) dx \\ &= 6\pi^2 - \left[2 \sin\left(\frac{x}{2}\right) \right]_{x=-2\pi}^{x=4\pi} \\ &= 6\pi^2 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the antiderivative

1 point is earned for the answer

5-2-

$$\begin{aligned}
 \int_{-2\pi}^{4\pi} f(x) dx &= \int_{-2\pi}^{4\pi} \left(g(x) - \cos\left(\frac{x}{2}\right) \right) dx \\
 &= 6\pi^2 - \left[2 \sin\left(\frac{x}{2}\right) \right]_{x=-2\pi}^{x=4\pi} \\
 &= 6\pi^2
 \end{aligned}$$

Part B

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{d}{dx}\left(\cos\left(\frac{x}{2}\right)\right)$

1 point is earned for $g'(x)$

1 point is earned for $x = 0$

1 point is earned for $x = \pi$

5-2-

$$f'(x) = g'(x) + \frac{1}{2}\sin\left(\frac{x}{2}\right) = \begin{cases} 1 + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } -2\pi < x < 0 \\ -\frac{1}{2} + \frac{1}{2}\sin\left(\frac{x}{2}\right) & \text{for } 0 < x < 4\pi \end{cases}$$

$f'(x)$ does not exist at $x = 0$.

For $-2\pi < x < 0$, $f'(x) \neq 0$.

For $0 < x < 4\pi$, $f'(x) = 0$ when $x = \pi$.

f has critical points at $x = 0$ and $x = \pi$.

Part C

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for $h'(x)$

1 point is earned for the answer

$$h'(x) = g(3x) \cdot 3$$

$$h'\left(-\frac{\pi}{3}\right) = 3g(-\pi) = 3\pi$$

13. If f is a continuous function on the closed interval $[a, b]$, which of the following must be true?

(A) There is a number c in the open interval (a, b) such that $f(c) = 0$.

(B) There is a number c in the open interval (a, b) such that $f(a) < f(c) < f(b)$.

(C) There is a number c in the closed interval $[a, b]$ such that $f(c) \geq f(x)$ for all x in $[a, b]$.



(D) There is a number c in the open interval (a, b) such that $f'(c) = 0$

(E) There is a number c in the open interval (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

5-2-

14. If f is continuous for $a \leq x \leq b$ and differentiable for $a < x < b$, which of the following could be false?

(A) $f'(c) = \frac{f(b)-f(a)}{b-a}$ for some c such that $a < c < b$.

(B) $f'(c)=0$ for some c such that $a < c < b$.



(C) f has a minimum value on $a \leq x \leq b$.

(D) f has a maximum value on $a \leq x \leq b$.

(E) $\int_a^b f(x)dx$ exists.

5-2-

15. Let f be the function defined by $f(x) = k\sqrt{x} - \ln x$, for $x > 0$, where k is a positive constant. For what value of the constant k does f have a critical point at $x=1$? For this value of k , determine whether f has a relative minimum, relative maximum, or neither at $x = 1$. Justify your answer.

Part B

1 point is earned for: sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned for solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

$$\text{When, } k = 2, f'(1) = 0, \text{ and, } f''(1) = -\frac{1}{2} + 1 > 0.$$

f has a relative minimum value at $x = 1$ by the Second Derivative Test.



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: sets $f'(1) = 0$ or $f'(x) = 0$

1 point is earned for solves for k

1 point is earned for the answer

1 point is earned for the justification

$$f'(1) = \frac{1}{2}k - 1 = 0 \Rightarrow k = 2$$

$$\text{When, } k = 2, f'(1) = 0, \text{ and, } f''(1) = -\frac{1}{2} + 1 > 0.$$

f has a relative minimum value at $x = 1$ by the Second Derivative Test.

16. The function f is defined for all x in the closed interval $[a, b]$. If f does not attain a maximum value on $[a, b]$, If f does not attain a maximum value on which of the following must be true?

(A) f is not continuous on $[a, b]$.

(B) f is not bounded on $[a, b]$.

(C) f does not attain a minimum value on $[a, b]$.

(D) The graph of f has a vertical asymptote in the interval $[a, b]$.

(E) The equation $f'(x) = 0$ does not have a solution in the interval $[a, b]$.



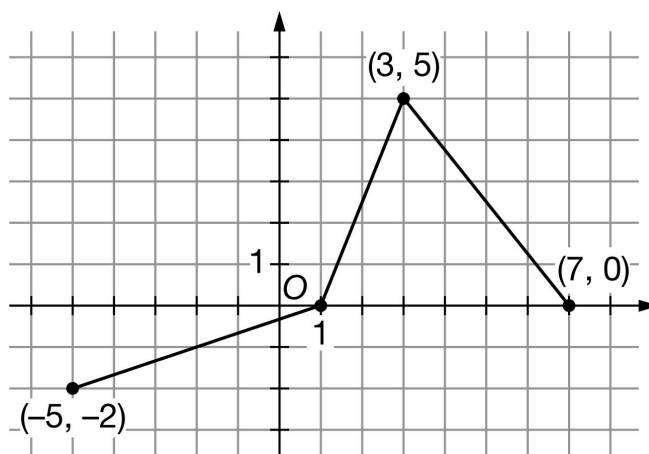
5-2-

17. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-5 \leq x \leq 7$ consists of three line segments, as shown in the figure above. It is known that $h(-5) = 6$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 7$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 7$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 1$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(3) = 5$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 1$ only

5-2-

- $x = 1$ and any additional x -coordinates not in $-5 < x < 7$

Does not earn the point:

- $x = 1$ and any additional x -coordinates in $-5 < x < 7$
- $x = 1$ with any incorrect supporting work

Do not read work related to additional values not in $-5 < x < 7$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$h'(x) = 0 \Rightarrow x = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 1$
- the slope of h changes from negative to positive at $x = 1$
- the graph changes from negative to positive at $x = 1$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 1$ and increasing to the right of $x = 1$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 1$

Note: If additional x -coordinates in $-5 < x < 7$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-5 < x < 7$ are identified in part (a), do not read these responses. Only read for values in the interval.

5-2-



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -5$ and $x = 7$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-5) = 6$ in calculation for $h(1)$ or $h(7)$

Third point: Answer with justification

The response must declare as an answer the absolute maximum value of 15.

The ordered pair $(7, 15)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-5	6
1	0
7	15



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

5-2-

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-5 < x < 7$ is a relative minimum, so the absolute maximum value of h on $-5 \leq x \leq 7$ must occur at an endpoint.

$$h(-5) = 6$$

$$\begin{aligned} h(7) &= h(-5) + \int_{-5}^7 h'(t) \, dt \\ &= 6 + \int_{-5}^1 h'(t) \, dt + \int_1^7 h'(t) \, dt \\ &= 6 - \frac{1}{2} \cdot 6 \cdot 2 + \frac{1}{2} \cdot 6 \cdot 5 = 15 \end{aligned}$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 7$ is $h(7) = 15$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from increasing to decreasing
- h'' changes from positive to negative at $x = 3$
- the slope of h' changes from positive to negative at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative maximum exists on the graph of h' at $x = 2$

Does not earn the point:

- the concavity changes at $x = 3$ [Must appeal to the change in sign of h'']
- h' changes from positive to negative
- the graph changes from positive to negative
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of h' at $x = 2$

5-2-

- the acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from increasing to decreasing there .

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 1$ is not the location of a point of inflection because

- h' is increasing for $x < 1$ and $x > 1$
- h' is increasing through $x = 1$
- h'' is always positive before and after $x = 1$
- the slope of h' is always positive before and after $x = 1$
- h'' does not change sign at $x = 1$
- a relative extrema does not exist on the graph of h' at $x = 1$

Does not earn the point:

- the concavity does not change [Must appeal to the change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

Answer with reason

5-2-

**Solution:**

$x = 1$ is not the location of a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (h'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{h'(x)}{2x}$$

$$\cdot \frac{h'(-2)}{2(-2)}$$

$$\cdot \frac{h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{12}$ or equivalent

5-2-



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$ and $\lim_{x \rightarrow -2} h''(x) = h''(-2)$.

$$\lim_{x \rightarrow -2} h'(x) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4} &= \lim_{x \rightarrow -2} \frac{h''(x)}{2x} \\ &= \frac{h''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{3}}{-4} = -\frac{1}{12} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

1st point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(3) = 3 \cdot h'(3) + h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the 1st point

Must be our answer, $g'(3) = 20$ or equivalent

5-2-



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(3) = 3 \cdot h'(3) + h(3) = 3 \cdot 5 + 5 = 20$$

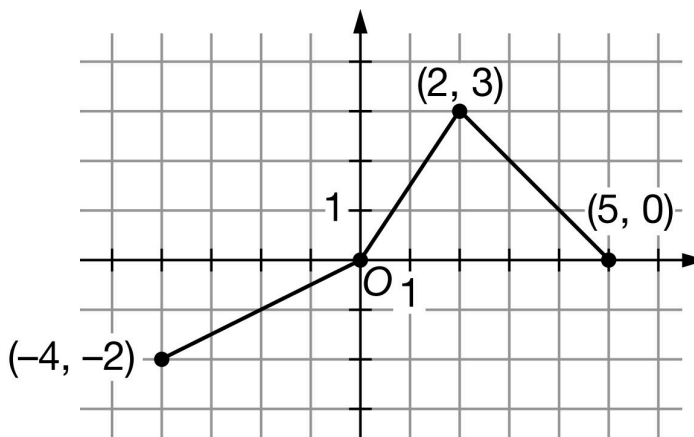
5-2-

18. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-4 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-4) = 7$.

- Find the x -coordinate of each critical point of g on the open interval $-4 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute maximum value of g on the interval $-4 \leq x \leq 5$. Justify your answer.
- Is $x = 2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = x \cdot g(x)$. It is known that $g(2) = 6$. Find the value of $h'(2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

5-2-

- $x = 0$ and any additional x -coordinates not in $-4 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-4 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-4 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $g'(x)$ changes from negative to positive at $x = 0$
- the slope of g changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- g is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-4 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is NOT eligible to earn the point.

Note: If additional x -coordinates not in $-4 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.

5-2-



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for g because $g'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -4$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-4) = 7$ in a calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 10.5.

The ordered pair $(5, 10.5)$ is not a declaration of the maximum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-4	7
0	3
6	10.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

5-2-

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-4 < x < 5$ is a relative minimum, so the absolute maximum value of g on $-4 \leq x \leq 5$ must occur at an endpoint.

$$g(-4) = 7$$

$$\begin{aligned} g(5) &= g(-4) + \int_{-4}^5 g'(t) \, dt \\ &= 7 + \int_{-4}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= 7 - \frac{1}{2} \cdot 4 \cdot 2 + \frac{1}{2} \cdot 3 \cdot 5 = 10.5 \end{aligned}$$

Therefore, the absolute maximum value of g on $-4 \leq x \leq 5$ is $g(5) = 10.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = 2$
- the slope of g' changes from positive to negative at $x = 2$
- g'' changes sign at $x = 2$
- a relative extrema exists on the graph of g' at $x = 2$
- a relative maximum exists on the graph of g' at $x = 2$

Does not earn the point:

- the concavity at $x = 2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ or $g''(x)$ does not exist (DNE)
- a relative minimum exists on the graph of g' at $x = 2$

5-2-

- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is increasing for $x < 0$ and $x > 0$
- g' is increasing through $x = 0$
- g'' is always positive before and after $x = 0$
- the slope of g' is always positive before and after $x = 0$
- g'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always positive
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

Answer with reason

5-2-

**Solution:**

$x = 0$ is not the location of a point of inflection for the graph of g because g' is increasing on both sides of $x = 0$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow -2} (g'(x) + 1) = 0 \text{ and } \lim_{x \rightarrow -2} (x^2 - 4) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -2} \frac{g'(x)}{2x}$$

$$\cdot \frac{g'(-2)}{2(-2)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $-\frac{1}{8}$ or equivalent

5-2-



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} g'(x) = g'(-2)$ and $\lim_{x \rightarrow -2} g''(x) = g''(-2)$.

$$\lim_{x \rightarrow -2} (g'(x) + 1) = g'(-2) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{g''(x)}{2x} \\ &= \frac{g''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{2}}{-4} = -\frac{1}{8} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product rule

Applies product rule: $h'(x) = x \cdot g'(x) + g(x)$

Product rule may appear as $h'(2) = 2 \cdot g'(2) + g(2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(2) = 12$ or equivalent

5-2-



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = x \cdot g'(x) + g(x)$$

$$h'(2) = 2 \cdot g'(2) + g(2) = 2 \cdot 3 + 6 = 12$$

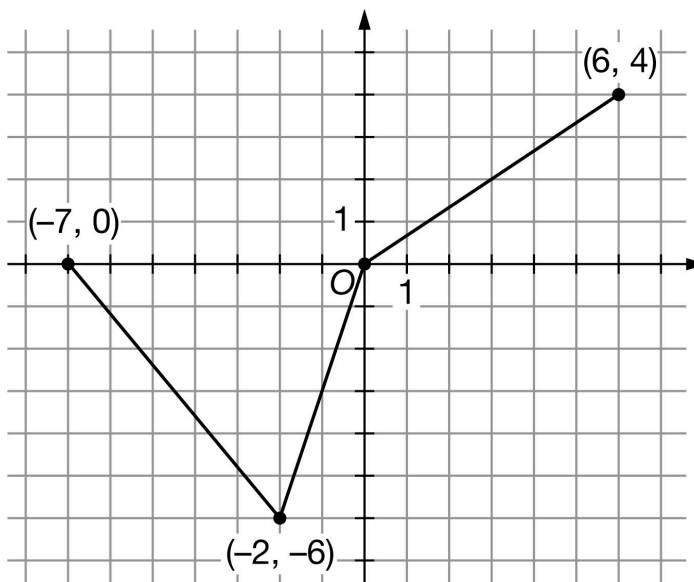
5-2-

19. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. It is known that $h(-7) = 2$.

- Find the x -coordinate of each critical point of h on the open interval $-7 < x < 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-7 \leq x \leq 6$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(-2) = -13$. Find the value of $g'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-2-

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-7 < x < 6$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-7 < x < 6$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-7 < x < 6$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 0$
- the slope of h changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$
- Any incorrect argument

5-2-

Note: If additional x -coordinates in $-7 < x < 6$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-7 < x < 6$ are identified in part (a), do not read reasons for these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -7$ and $x = 6$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-7) = 2$ in a calculation for $h(0)$ and/or $h(6)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 2.

The ordered pair $(-7, 2)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-7	2
0	-19
6	-7

5-2-



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-7 < x < 6$ is a relative minimum, so the absolute maximum value of h on $-7 \leq x \leq 6$ must occur at an endpoint.

$$h(-7) = 2$$

$$\begin{aligned} h(6) &= h(-7) + \int_{-7}^6 h'(t) dt \\ &= 2 + \int_{-7}^0 h'(t) dt + \int_0^6 h'(t) dt \\ &= 2 - \frac{1}{2} \cdot 7 \cdot 6 + \frac{1}{2} \cdot 6 \cdot 4 = -7 \end{aligned}$$

Therefore, the absolute maximum value of h on $-7 \leq x \leq 6$ is $h(-7) = 2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' changes from negative to positive at $x = -2$
- the slope of h' changes from negative to positive at $x = -2$
- h'' changes sign at $x = -2$
- a relative extrema exists on the graph of h' at $x = -2$
- a relative minimum exists on the graph of h' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of h'']

5-2-

- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = -2$
- The acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is increasing for $x < 0$ and $x > 0$
- h' is increasing through $x = 0$
- h'' is always positive before and after $x = 0$
- the slope of h' is always positive before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of h'']
- h' is always positive
- h' does not change sign

5-2-



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 0$ is not the location a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow 3} (2h'(x) - 4) = 0 \text{ and } \lim_{x \rightarrow 3} (x^2 - 9) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} \rightarrow 0$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$$

$$\text{Does not earn the point in the presence of limit } = \frac{0}{0}, \text{ i.e., } \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \frac{0}{0}$$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x)}{2x}$$

$$\cdot \frac{2h''(3)}{2(3)}$$

$$\cdot \frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

5-2-

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{2}{9}$ or equivalent

0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:
 h' and h'' are continuous at $x = 3$. $\Rightarrow \lim_{x \rightarrow 3} h'(x) = h'(3)$ and $\lim_{x \rightarrow 3} h''(x) = h''(3)$.

$$\lim_{x \rightarrow 3} 2h'(x) - 4 = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \lim_{x \rightarrow 3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(3)}{3} = \frac{\frac{2}{3}}{3} = \frac{2}{9}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$ Product rule may appear as $g'(-2) = -2 \cdot h'(-2) + h(-2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

5-2-

Eligibility: Earns the first point

Must be our answer, $g'(-2) = -1$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(-2) = -2 \cdot h'(-2) + h(-2) = -2 \cdot (-6) + (-13) = -1$$

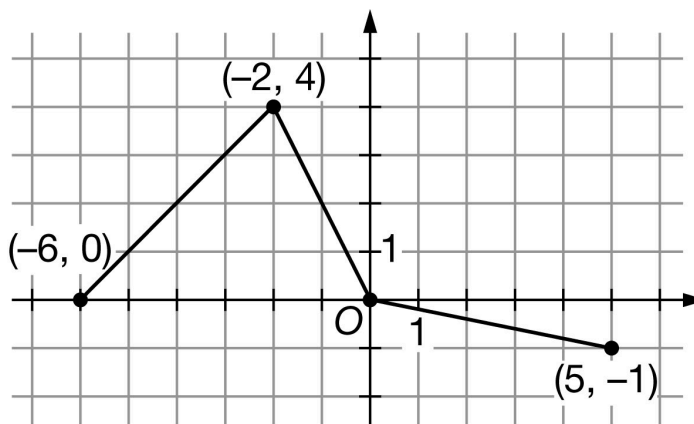
5-2-

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-6) = -1$.

- Find the x -coordinate of each critical point of g on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute minimum value of g on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -1} \frac{g'(x)-2}{x^2-1}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = 3x \cdot g(x)$. It is known that $g(-2) = 7$. Find the value of $h'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

5-2-

- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $g'(x)$ changes from positive to negative at $x = 0$
- the slope of g changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- g is increasing to the left of $x = 0$ and decreasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval

5-2-



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for g because $g'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-6) = -1$ in calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the absolute minimum value of -1 .

The ordered pair $(-6, -1)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-6	-1
0	11
5	8.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

☐ Considers both endpoints

5-2-

- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of g on $-6 \leq x \leq 5$ must occur at an endpoint.

$$g(-6) = -1$$

$$\begin{aligned} g(5) &= g(-6) + \int_{-6}^5 g'(t) dt \\ &= -1 + \int_{-6}^0 g'(t) dt + \int_0^5 g'(t) dt \\ &= -1 + \frac{1}{2} \cdot 6 \cdot 4 - \frac{1}{2} \cdot 5 \cdot 1 = \frac{17}{2} \end{aligned}$$

Therefore, the absolute minimum value of g on $-6 \leq x \leq 5$ is $g(-6) = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = -2$
- the slope of g' changes from positive to negative at $x = -2$
- g'' changes sign at $x = -2$
- a relative extrema exists on the graph of g' at $x = -2$
- a relative maximum exists on the graph of g' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ OR $g''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of g' at $x = -2$

5-2-

- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is decreasing for $x < 0$ and $x > 0$
- g' is decreasing through $x = 0$
- g'' is always negative before and after $x = 0$
- the slope of g' is always negative before and after $x = 0$
- g'' does not change sign at $x = 0$.
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always negative
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

5-2-

☐ Answer with reason
Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -1$

Earns the point:

$$\cdot \lim_{x \rightarrow -1} (g'(x) - 2) = 0 \text{ and } \lim_{x \rightarrow -1} (x^2 - 1) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -1} \frac{g'(x)}{2x}$$

$$\cdot \frac{g''(-1)}{2(-1)}$$

$$\cdot \frac{g'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer 1 or equivalent

5-2-



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -1$. $\Rightarrow \lim_{x \rightarrow -1} g'(x) = g'(-1)$ and $\lim_{x \rightarrow -1} g''(x) = g''(-1)$.

$$\lim_{x \rightarrow -1} g'(x) - 2 = 2 - 2 = 0$$

$$\lim_{x \rightarrow -1} (x^2 - 1) = 1^2 - 1 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{g''(x)}{2x} \\ &= \frac{g''(-1)}{2 \cdot (-1)} = \frac{-2}{-2} = 1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $h'(x) = 3x \cdot g'(x) + 3 \cdot g(x)$

Product rule may appear as $h'(-2) = -6 \cdot g'(-2) + 3 \cdot g(-2)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(-2) = -3$ or equivalent

5-2-



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$h'(x) = 3x \cdot g'(x) + 3g(x)$$

$$h'(-2) = 3 \cdot (-2) \cdot g'(-2) + 3g(-2) = -6 \cdot 4 + 3 \cdot 7 = -3$$

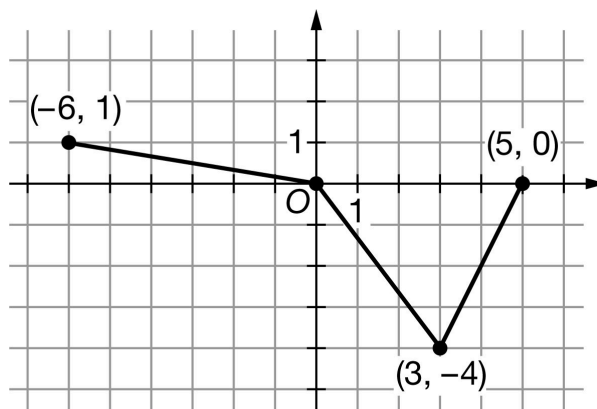
5-2-

21. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $h(-6) = 5$.

- Find the x -coordinate of each critical point of h on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute minimum value of h on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -3} \frac{2h'(x)-1}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = 2x \cdot h(x)$. It is known that $h(3) = 2$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

· $x = 0$ only

5-2-

- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$

- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer
Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $h'(x)$ changes from positive to negative at $x = 0$
- the slope of h changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- h is increasing to the left of $x = 0$
- and decreasing to the right of $x = 0$
- (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to

5-2-

earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for h because $h'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the 1st point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-6) = 5$ in a calculation for $h(0)$ or $h(5)$

Third point: Answer with justification

The response must declare as an answer the minimum value of -2 .

The ordered pair $(5, -2)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-6	5
0	8
5	-2



0	1	2	3
---	---	---	---

5-2-

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of h on $-6 \leq x \leq 5$ must occur at an endpoint.

$$h(-6) = 5$$

$$\begin{aligned} h(5) &= h(-6) + \int_{-6}^5 h'(t) dt \\ &= 5 + \int_{-6}^0 h'(t) dt + \int_0^5 h'(t) dt \\ &= 5 + \frac{1}{2} \cdot 6 \cdot 1 - \frac{1}{2} \cdot 5 \cdot 4 = -2 \end{aligned}$$

Therefore, the absolute minimum value of h on $-6 \leq x \leq 5$ is $h(5) = -2$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' shifts from negative to positive at $x = 3$
- the slope of h' shifts from negative to positive at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative minimum exists on the graph of h' at $x = 3$

Does not earn the point:

- the concavity at $x = 3$ changes. [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive

5-2-

- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = 3$
- the acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is decreasing for $x < 0$ and $x > 0$
- h' is decreasing through $x = 0$
- h'' is always negative before and after $x = 0$
- the slope of h' is always negative before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change. [Must appeal to the lack of change in sign of h'']
- h' is always negative
- h' does not change sign



0	1
---	---

5-2-

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of h because h' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -3$

Earns the point:

$$\cdot \lim_{x \rightarrow -3} (2h'(x) - 1) = 0 \text{ and } \lim_{x \rightarrow -3} (x^2 - 9) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} \rightarrow 0$$

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow -3} \frac{2h'(x)}{2x}$$

$$\cdot \frac{2h'(-3)}{2(-3)}$$

$$\cdot \frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

5-2-

Must be our answer $\frac{1}{18}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -3$. $\Rightarrow \lim_{x \rightarrow -3} h'(x) = h'(-3)$ and $\lim_{x \rightarrow -3} h''(x) = h''(-3)$.

$$\lim_{x \rightarrow -3} 2h'(x) - 1 = 2 \cdot \frac{1}{2} - 1 = 0$$

$$\lim_{x \rightarrow -3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \lim_{x \rightarrow -3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(-3)}{-3} = \frac{-\frac{1}{6}}{-3} = \frac{1}{18}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

Applies product rule: $g'(x) = 2x \cdot h'(x) + 2 \cdot h(x)$

Product rule may appear as $g'(3) = 6 \cdot h'(3) + 2 \cdot h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

5-2-

Must be our answer, $g'(3) = -20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Product rule

☐ Answer

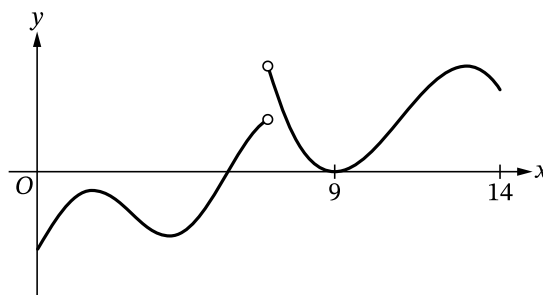
Solution:

$$g'(x) = 2x \cdot h'(x) + 2h(x)$$

$$g'(3) = 2 \cdot 3 \cdot h'(3) + 2h(3) = 6 \cdot (-4) + 2 \cdot 2 = -20$$

5-2-

22. The function f is continuous on the interval $0 < x < 14$. Shown in the figure is the graph of f' , the derivative of f , on the interval $0 < x < 14$. The graph of f' is tangent to the x -axis at $x = 9$.

Graph of f'

How many critical points does the function f have on the interval $0 < x < 14$?

- (A) One
(B) Two
(C) Three
(D) Four



Answer C

Correct. A critical point for the function f is where the derivative f' does not exist or where $f'(x) = 0$. According to the graph of the derivative, $f'(x) = 0$ at $x = 9$ and at one other value of x less than 9, and f' does not exist at one value of x that is between the two zeros. Therefore the function f has three critical points on the interval $0 < x < 14$.

5-2-

23. Let f be the function given by $f(x) = \frac{\sin x \cos x}{(x^2 - 4)}$ on the closed interval $[-2\pi, 2\pi]$. On which of the following closed intervals is the function f guaranteed by the Extreme Value Theorem to have an absolute maximum and an absolute minimum?
- (A) $[-2\pi, 2\pi]$
- (B) $[-2, 2]$
- (C) $[-1, 1]$ ✓
- (D) $[\frac{\pi}{2}, \pi]$

Answer C

Correct. The Extreme Value Theorem guarantees that a function that is continuous on a closed interval has both an absolute maximum and an absolute minimum on that closed interval. The rational function f is continuous on the closed interval $[-1, 1]$ because this interval does not contain $x = -2$ or $x = 2$, where f is undefined.

24. Let f be the function given by $f(x) = \frac{x+4}{(x-1)(x+3)}$ on the closed interval $[-5, 5]$. On which of the following closed intervals is the function f guaranteed by the Extreme Value Theorem to have an absolute maximum and an absolute minimum?
- (A) $[-5, 5]$
- (B) $[-3, 1]$
- (C) $[-2, 0]$ ✓
- (D) $[0, 5]$

Answer C

Correct. The Extreme Value Theorem guarantees that a function that is continuous on a closed interval has both an absolute maximum and an absolute minimum on that closed interval. The rational function f is continuous on the closed interval $[-2, 0]$ because this interval does not contain $x = 1$ or $x = -3$, where f is undefined.

5-2-

25. Let f be the function defined by $f(x) = x \sin x$ with domain $[0, \infty)$. The function f has no absolute minimum and no absolute maximum on its domain. Why does this not contradict the Extreme Value Theorem?
- (A) The domain of f is not an open interval.
- (B) The domain of f is not a closed and bounded interval. ✓
- (C) The function f is not continuous on its domain.
- (D) The function f is not differentiable on its domain.

Answer B

Correct. The Extreme Value Theorem applies to continuous functions over closed intervals of the form $[a, b]$, where $a < b$. Although the function f is continuous on its domain, the interval $[0, \infty)$ is not closed and bounded. Therefore the Extreme Value Theorem does not apply.

26. Which of the following functions of x is guaranteed by the Extreme Value Theorem to have an absolute maximum on the interval $[0, 4]$?
- (A) $y = \tan x$
- (B) $y = \tan^{-1} x$ ✓
- (C) $y = \frac{x^2 - 16}{x^2 + x - 20}$
- (D) $y = \frac{1}{e^x - 1}$

Answer B

Correct. The Extreme Value Theorem guarantees that a function that is continuous on a closed interval has an absolute maximum on that closed interval. The function $y = \tan^{-1} x$ is continuous for all x , and therefore it satisfies the conditions for the Extreme Value Theorem on the closed interval $[0, 4]$.

5-2-

27. Let f be the function defined by $f(x) = \frac{x^2+1}{x+1}$ with domain $[0, \infty)$. The function f has no absolute maximum on its domain. Why does this not contradict the Extreme Value Theorem?
- (A) The domain of f is not an open interval.
- (B) The domain of f is not a closed and bounded interval. ✓
- (C) The function f is not continuous on its domain.
- (D) The function f is not differentiable on its domain.

Answer B

Correct. The Extreme Value Theorem applies to continuous functions over closed intervals of the form $[a, b]$, where $a < b$. Although the function f is continuous on its domain, the interval $[0, \infty)$ is not closed and bounded. Therefore, the Extreme Value Theorem does not apply.

28. Which of the following functions of x is guaranteed by the Extreme Value Theorem to have an absolute maximum on the interval $[0, 2\pi]$?
- (A) $y = \frac{1}{1+\sin x}$
- (B) $y = \frac{1}{x^2+\pi}$ ✓
- (C) $y = \frac{x^2-2\pi x+\pi^2}{x-\pi}$
- (D) $y = \frac{|x-\pi|}{x-\pi}$

Answer B

Correct. The Extreme Value Theorem guarantees that a function that is continuous on a closed interval has an absolute maximum on that closed interval. The function $y = \frac{1}{x^2+\pi}$ is continuous for all x , and therefore it satisfies the conditions for the Extreme Value Theorem on the closed interval $[0, 2\pi]$.

5-2-

29.

x	0	1	2	3
$f(x)$	0	4	7	6

Let f be a function with selected values given in the table above. Which of the following statements must be true?

- I. By the Intermediate Value Theorem, there is a value c in the interval $(0, 3)$ such that $f(c) = 2$.
- II. By the Mean Value Theorem, there is a value c in the interval $(0, 3)$ such that $f'(c) = 2$.
- III. By the Extreme Value Theorem, there is a value c in the interval $[0, 3]$ such that $f(c) \leq f(x)$ for all x in the interval $[0, 3]$.

(A) None



(B) I only

(C) II only

(D) I, II, and III

Answer A

Correct. If f were continuous on the closed interval $[0, 3]$, then statement I would follow from the Intermediate Value Theorem, statement III would follow from the Extreme Value Theorem with f having an absolute minimum on the interval $[0, 3]$, and statement II would follow from the Mean Value Theorem if f were also differentiable on the open interval $(0, 3)$. However, no assumption was given about f being continuous, so none of these statements must be true.

5-2-

30. Let g be the function given by $g(x) = \sqrt{1 - \sin^2 x}$. Which of the following statements could be false on the interval $0 \leq x \leq \pi$?
- (A) By the Extreme Value Theorem, there is a value c such that $g(c) \leq g(x)$ for $0 \leq x \leq \pi$.
- (B) By the Extreme Value Theorem, there is a value c such that $g(c) \geq g(x)$ for $0 \leq x \leq \pi$.
- (C) By the Intermediate Value Theorem, there is a value c such that $g(c) = \frac{g(0)+g(\pi)}{2}$.
- (D) By the Mean Value Theorem, there is a value c such that $g'(c) = \frac{g(\pi)-g(0)}{\pi-0}$. ✓

Answer D

Correct. This is an attempt to use the Mean Value Theorem to conclude that there is a c for which $g'(c) = \frac{g(b)-g(a)}{b-a}$ on the interval $[a, b]$. However, the theorem requires that g be differentiable on the open interval $(0, \pi)$. The function g is not differentiable on this interval because the derivative $g'(x) = -\frac{\sin x \cos x}{\sqrt{1-\sin^2 x}}$ does not exist at $x = \frac{\pi}{2}$.

5-2-

31.

x	0	1	2	3
$f(x)$	15	14	12	9

Let f be a function with selected values given in the table above. Which of the following statements must be true?

- I. By the Intermediate Value Theorem, there is a value c in the interval $(0, 3)$ such that $f(c) = 10$.
- II. By the Mean Value Theorem, there is a value c in the interval $(0, 3)$ such that $f'(c) = -2$.
- III. By the Extreme Value Theorem, there is a value c in the interval $[0, 3]$ such that $f(c) \leq f(x)$ for all x in the interval $[0, 3]$.

(A) None



(B) I only

(C) II only

(D) I, II, and III

Answer A

Correct. If f were continuous on the closed interval $[0, 3]$, then statement I would follow from the Intermediate Value Theorem, statement III would follow from the Extreme Value Theorem with f having an absolute minimum on the interval $[0, 3]$, and statement II would follow from the Mean Value Theorem if f were also differentiable on the open interval $(0, 3)$. However, no assumption was given about f being continuous, so none of these statements must be true.

5-2-

32.

x	2	3	4	5
$f(x)$	1	14	20	31

Selected values of a continuous function f are given in the table above. Which of the following statements could be false?

- (A) By the Intermediate Value Theorem applied to f on the interval $[2, 5]$, there is a value c such that $f(c) = 10$.
- (B) By the Mean Value Theorem applied to f on the interval $[2, 5]$, there is a value c such that $f'(c) = 10$. ✓
- (C) By the Extreme Value Theorem applied to f on the interval $[2, 5]$, there is a value c such that $f(c) \leq f(x)$ for all x in $[2, 5]$.
- (D) By the Extreme Value Theorem applied to f on the interval $[2, 5]$, there is a value c such that $f(c) \geq f(x)$ for all x in $[2, 5]$.

Answer B

Correct. This is an attempt to use the Mean Value Theorem to conclude that there is a value c for which $f'(c) = \frac{f(5)-f(2)}{5-2} = \frac{30}{3} = 10$. However, that theorem requires that the function f be differentiable on the open interval $(2, 5)$. The only condition known about f is that it is continuous, not that it is differentiable, so this statement could be false. For example, if f is piecewise linear on the intervals $(-\infty, 3]$, $[3, 4]$, and $[4, \infty)$, there would be no value c where $f'(c) = 10$.

5-2-

33. Let g be the function given by $g(x) = \sqrt{1 + \cos x}$. Which of the following statements could be false on the interval $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$?
- (A) By the Extreme Value Theorem, there is a value c such that $g(c) \leq g(x)$ for $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$.
- (B) By the Extreme Value Theorem, there is a value c such that $g(c) \geq g(x)$ for $\frac{\pi}{2} \leq x \leq \frac{7\pi}{4}$.
- (C) By the Intermediate Value Theorem, there is a value c such that $g(c) = \frac{g(\frac{\pi}{2}) + g(\frac{7\pi}{4})}{2}$.
- (D) By the Mean Value Theorem, there is a value c such that $g'(c) = \frac{g(\frac{7\pi}{4}) - g(\frac{\pi}{2})}{\frac{7\pi}{4} - \frac{\pi}{2}}$. ✓

Answer D

Correct. This is an attempt to use the Mean Value Theorem to conclude that there is a c for which $g'(c) = \frac{g(b) - g(a)}{b - a}$ on the interval $[a, b]$. However, the theorem requires that g be differentiable on the open interval $(\frac{\pi}{2}, \frac{7\pi}{4})$. The function g is not differentiable on this interval because the derivative $g'(x) = \frac{-\sin x}{2\sqrt{1 + \cos x}}$ does not exist at $x = \pi$. There is no guarantee that a solution exists in this interval. (No such c exists in this case.)

5-2-

34.

x	0	1	2	3	4
$f(x)$	2	-2	4	7	18

Selected values of a continuous function f are given in the table above. Which of the following statements could be false?

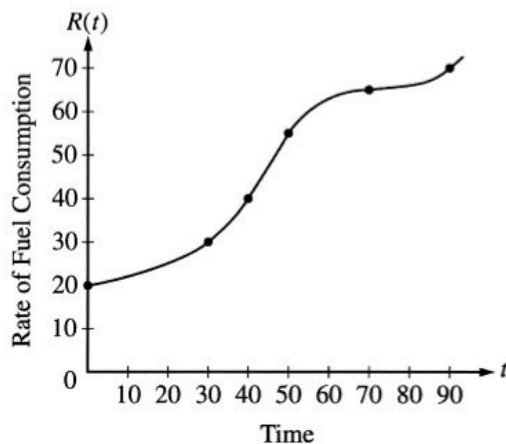
- (A) By the Intermediate Value Theorem applied to f on the interval $[0, 4]$, there is a value c such that $f(c) = 5$.
- (B) By the Mean Value Theorem applied to f on the interval $[0, 4]$, there is a value c such that $f'(c) = 4$. ✓
- (C) By the Extreme Value Theorem applied to f on the interval $[0, 4]$, there is a value c such that $f(c) \leq f(x)$ for all x in $[0, 4]$.
- (D) By the Extreme Value Theorem applied to f on the interval $[0, 4]$, there is a value c such that $f(c) \geq f(x)$ for all x in $[0, 4]$.

Answer B

Correct. This is an attempt to use the Mean Value Theorem to conclude that there is a value c for which $f'(c) = \frac{f(4)-f(0)}{4-0} = \frac{18-2}{4} = 4$. The Mean Value Theorem requires that f is continuous on the closed interval $[0, 4]$ and differentiable on the open interval $(0, 4)$. Because we do not know that f is differentiable on the open interval $(0, 4)$, the Mean Value Theorem does not apply.

5-2-

35.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

(a) Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

(b) The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

(c) Approximate the value of $\int_0^{90} R(t) dt$ using a left Riemann sum with the five subintervals indicated by the data in the table. Is this numerical approximation less than the value of $\int_0^{90} R(t) dt$? Explain your reasoning.

(d) For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane. Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units



0	1	2
---	---	---

5-2-

The student response earns all of the following points:

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

Part C

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.



0	1	2
---	---	---

5-2-

The student response earns all of the following points:

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t)dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.

Part D

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

5-2-

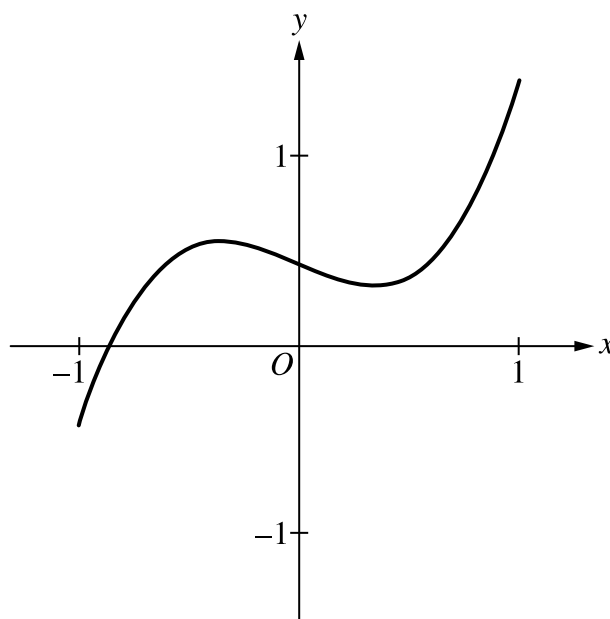
1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t) dt$

$\frac{1}{b} \int_0^b R(t) dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers

36.



Graph of f

The figure shows the graph of the differentiable function f for $-1 < x < 1$. How many critical points does the graph of $y = (f(x))^2$ have on the interval given?

- (A) One
- (B) Two
- (C) Three
- (D) Four



Answer C

Correct. By the chain rule, $\frac{dy}{dx} = \frac{d}{dx} (f(x))^2 = 2f(x)f'(x)$. There is one solution to $2f(x)f'(x) = 0$ where $f(x) = 0$. There are two more solutions where $f'(x) = 0$ at the relative extrema on the graph of f . There are no points where the derivative is undefined. Therefore the graph of $y = (f(x))^2$ has a total of three critical points.